TEST PLAN SPECIFICATION

SimpleShell — Command-Line Interpreter

Document Version 2.1 | Project #18 | Language: C11 | Platform: Linux / POSIX

Project Team Members

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1. Purpose and scope

This plan defines verification activities for SimpleShell. It follows the requirement groups and acceptance criteria referenced by the supplied project documents: FR-01 through FR-08, NFR-01 through NFR-12, and AC-01 through AC-10. Test execution must confirm each feature against the approved SRS and the implementation; no result is claimed as passed until it has actually been run.

2. Features in scope

Interactive REPL and EOF handling; whitespace and quoted-token parsing; built-ins cd, pwd, echo, export, unset, help, and exit; external command lookup and process execution; input/output/append/stderr redirection; pipelines; signal behaviour; environment and status expansion; error handling, portability, and memory checks.

3. Features out of scope for v1.0

Command history, interactive tab completion, shell scripting control structures, background job-control commands (bg/fg/jobs), and native Windows Win32 execution. The intended environment is POSIX/Linux or a POSIX-compatible environment such as WSL2.

4. Test approach and environment

Use automated non-interactive tests for deterministic inputs and outputs, manual terminal tests for Ctrl-C/Ctrl-Z, boundary tests for long input and chained pipelines, compiler diagnostics, and Valgrind where available. Environment: Ubuntu 22.04 LTS (primary), compatible Linux/WSL2; GCC 11+ or compatible C11 compiler; GNU Make; POSIX utilities; Valgrind 3.18+ for memory checks.

Build command specified by the existing project plan:

```
make clean && make
```

The plan expects the build to produce ./shell. Confirm this against the repository Makefile before running the tests.

5. Test case matrix

Test ID	SRS Ref	Scenario	Input / Action	Expected result / pass condition
TC-REPL-01	FR-01	Prompt display	Launch ./shell	A prompt is displayed and input is accepted.
TC-REPL-02	FR-01	EOF at empty prompt	Send Ctrl-D at an empty prompt	Shell exits cleanly with the SRS-defined status.
TC-REPL-03	FR-01	Blank input	Press Enter or enter whitespace	No command is executed; shell returns to prompt.
TC-PARSE-01	FR-02	Whitespace tokenization	echo hello world	Arguments are separated correctly; output is hello world.
TC-PARSE-02	FR-02	Single quotes	echo 'literal $VAR'	Quoted content remains literal according to SRS quote rules.
TC-PARSE-03	FR-02	Double quotes	Set A=X; echo "$A y"	Variable expansion and spaces follow the SRS rules.
TC-PARSE-04	FR-02	Token boundary	Provide at least 256 tokens if required by SRS	Input is handled without truncation or memory error.
TC-PARSE-05	FR-02	Unmatched quote	Enter an unterminated quoted string	Input is rejected safely with a diagnostic; shell remains usable.
TC-CMD-01	FR-04	PATH command	ls -la	External command is located and executed.
TC-CMD-02	FR-04	Child lifecycle	uname -s	Child executes and parent collects its termination status.
TC-CMD-03	FR-04	Unknown command	xyz_invalid_prog_123	Diagnostic is written to stderr; shell continues.
TC-CMD-04	NFR-11	PATH safety	Try a local executable when current directory is not explicitly in PATH	Lookup follows the SRS security rule; current directory is not implicitly trusted if prohibited.
TC-BUILTIN-01	FR-03	cd to directory	cd /tmp; pwd	Working directory changes and pwd reports it.
TC-BUILTIN-02	FR-03	cd default	Run cd without arguments	Changes to HOME if required by SRS.
TC-BUILTIN-03	FR-03	cd invalid directory	cd /missing_dir_xyz	Error is reported and shell continues.
TC-BUILTIN-04	FR-03	pwd	pwd	Prints current absolute working directory.
TC-BUILTIN-05	FR-03	echo	echo Hello World	Prints arguments in the specified format.
TC-BUILTIN-06	FR-03	export and unset	export K=V; unset K	Variable is set and then removed.
TC-BUILTIN-07	FR-03	help	help	Displays supported built-ins and usage summary.
TC-BUILTIN-08	FR-03	exit status	exit 42	Shell terminates using the requested exit status.
TC-REDIR-01	FR-05	Output redirect	echo hello > out.txt	File is created/truncated and receives stdout.
TC-REDIR-02	FR-05	Append redirect	echo next >> out.txt	Output is appended rather than replacing existing content.
TC-REDIR-03	FR-05	Input redirect	cat < input.txt	Command receives file content on stdin.
TC-REDIR-04	FR-05	Stderr redirect	Run a command that errors with 2> err.txt	Error output is redirected to err.txt.
TC-REDIR-05	FR-05	Combined redirects	cmd < in.txt > out.txt 2> err.txt	Supported redirections are applied without descriptor leakage.

Test ID	SRS Ref	Scenario	Input / Action	Expected result / pass condition
TC-PIPE-01	FR-06	Two-stage pipeline	ls | grep .c	Output of the first command becomes input to the second.
TC-PIPE-02	FR-06	Concurrent pipeline	sleep 1 | sleep 1 (or equivalent valid pipeline)	Pipeline stages do not run as a serial command chain.
TC-PIPE-03	FR-06	Chained pipeline	cat input.txt | sort | uniq | wc -l	Three or more stages complete without deadlock.
TC-SIG-01	FR-07	Ctrl-C during child	Run sleep 10, press Ctrl-C	Foreground child is interrupted; shell remains available.
TC-SIG-02	FR-07	Ctrl-C at prompt	Press Ctrl-C with no child running	Shell remains alive and prompt is restored according to SRS.
TC-SIG-03	FR-07	Ctrl-Z behaviour	Run sleep 10, press Ctrl-Z	Observed behaviour matches the SRS; do not claim job-control support if excluded.
TC-VAR-01	FR-08	Environment variable	echo $USER and echo ${USER}	Supported variable forms expand according to SRS.
TC-VAR-02	FR-08	Last status	Run a failing command, then echo $?	Prints the previous command status.
TC-VAR-03	FR-08	Shell PID	echo $$	Prints the shell process ID.
TC-QUAL-01	NFR-02	Memory check	Run a representative test set under Valgrind	No definite/indirect leaks are reported, if the tool is available.
TC-QUAL-02	NFR-08	Long input	Submit a 4096-character input line	Handled safely according to SRS limits; no crash or overflow.
TC-QUAL-03	NFR-01	Startup latency	Measure shell startup repeatedly	Compare measured latency with the threshold stated in the SRS.
TC-QUAL-04	NFR-04	Portability/build	Build with C11 warning flags on supported POSIX environment	Build conforms to documented portability and compiler requirements.

6. Execution, pass/fail, and traceability

Record actual result (Pass/Fail/Blocked), test date, environment, evidence, and defect reference for every test. A test is Pass only when observed behaviour matches the SRS and the expected result. Do not state that tests, CI runs, or Valgrind checks have run unless there is execution evidence.

Build and memory-check examples (adapt paths to the actual repository):

```
gcc -Wall -Wextra -Wpedantic -std=c11 ...
valgrind --leak-check=full --show-leak-kinds=all --error-exitcode=1 ./shell
```

7. Requirements Traceability Matrix

SRS requirement group	Scope	Verifying tests
FR-01	REPL, prompt, input line and EOF	TC-REPL-01 to TC-REPL-03
FR-02	Lexing/parsing, whitespace, quotes and token bounds	TC-PARSE-01 to TC-PARSE-05
FR-03	Built-in commands	TC-BUILTIN-01 to TC-BUILTIN-08
FR-04	External command search/execution and status	TC-CMD-01 to TC-CMD-03
FR-05	Input, output, append and stderr redirection	TC-REDIR-01 to TC-REDIR-05
FR-06	Pipelines	TC-PIPE-01 to TC-PIPE-03
FR-07	Interactive signal behaviour	TC-SIG-01 to TC-SIG-03
FR-08	Environment/status/PID expansion	TC-VAR-01 to TC-VAR-03
NFR-01	Startup performance threshold	TC-QUAL-03
NFR-02	Memory safety / leak checking	TC-QUAL-01
NFR-04	Portability/build compatibility	TC-QUAL-04
NFR-08	Long-input robustness	TC-QUAL-02
NFR-11	PATH security rule	TC-CMD-04

Acceptance criteria AC-01 through AC-10 must be mapped to their exact wording in the approved SRS. The uploaded item in this conversation was a GitHub HTML page rather than a readable DOCX, so this version preserves the requirement-group references visible in the existing documents; verify individual AC mappings against the actual SRS before final submission.